Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/30/2025
PI Name:	Prejean, Brian J. Ph.D.		
Project Title:	Efficacy of Orion Flywheel Exercise Training in Ambulatory Subjects		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Aerobic**:Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity (2) **Muscle**:Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/01/2024	**End Date:**	09/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI), Brian Crucian, Ph.D. has been added a Co-Investigator. Dr. Crucian is at NASA Johnson Space Center (Ed., 11/27/24).		
COI Name (Institution):	Amonette, William Ph.D. (University of Houston) Chappell, Lori M.S. (KBR/NASA Johnson Space Center) Marshall-Goebel, Karina Ph.D. (NASA Johnson Space Center) Strock, Nicole Ph.D. (KBR/NASA Johnson Space Center) Varanoske, Alyssa Ph.D. (KBR/NASA Johnson Space Center) Crucian, Brian (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Per the Principal Investigator (PI), the Task Description has changed. There has been a large rescoping of the study design. Specifically, the study arms were reduced from 3 exercising cohorts and 1 control cohort to a single Orion Flywheel exercise group. The majority of the originally proposed fitness and task performance measures were preserved at pre-, mid-, and post-participation timepoints, although blood biomarkers were largely reduced to those focused on inflammation and muscle damage. [Ed., 1/22/26). The Orion Flywheel (OFW) exercise device was selected to provide both resistance and aerobic (rowing ergometry) exercise for early Artemis missions to the Moon, lasting up to 21 days – which did not entail lunar surface extravehicular activities (EVAs). However, planned missions that will rely on the OFW as the sole exercise device have been extended to beyond 30 days in duration and will include lunar surface EVAs, which increase the performance demands on the crewmembers, and therefore the importance of the exercise device to protect against spaceflight-related deconditioning. The OFW provides speed-dependent exercise loads (known as isoinertial resistance) that are distinct from exercise loads provided by any other device that has been used in spaceflight, and only allows for a small number of resistance exercises. Other spaceflight exercise devices are designed to provide exercise loads that are comparable to well-studied, conventional gym-based exercise equipment (i.e., free weights, commercial cycles, rowing ergometers, and treadmills) and allow for a comprehensive range of resistance exercises. Additionally, the volume of OFW exercise that will be required of crewmembers on long-duration Artemis missions (30-60 minutes/day, 6 days/week) will exceed that used in previous ground-based flywheel studies. Therefore, understanding the efficacy of the novel OFW exercise device to be used as a comprehensive, stand-alone system will inform upcoming missing risk assessments and define future spaceflight exercise system designs. NASA exercise scientists will collaborate with researchers from the University of Houston-Clear Lake to execute a controlled exercise intervention in free-living subjects to characterize the effects of an 8-week training regimen following Artemis exercise time requirements using only the OFW. A distribution of subject anthropometries and fitness levels will be recruited, using bounding inclusion criteria to mimic the NASA astronaut population to properly assess the ability of the OFW to be useful and effective for all crewmembers. Adaptations in functional performance, aerobic fitness, muscle strength and endurance, body composition, as well as metabolic and inflammatory responses will be assessed pre-intervention and at the mid-point and end of the 8-week exercise regimen. Additionally, subjective measures such as exercise tolerance and satisfaction with the device will be collected across participation to further inform future mission operations. The investigators hypothesize that OFW-only exercise across an 8-week intervention will result in improvements across all categories of health and performance assessed. However, it is expected that limitations to the available exercises on OFW will be observed as reduced benefits or no change in upper body strength as well as functional tasks requiring pressing or overhead strength. Additionally, investigators expect to observe increases in muscle damage markers and inflammatory responses immediately post-exercise along with subjective ratings of muscle soreness or discomfort in the early weeks of the intervention. However, it is expected that the ability of exercisers to self-regulate their workloads with the Flywheel's isoinertial loading mechanism will allow subjects to become acclimated while continuing daily exercise, although some prescription adjustments may be needed. Exercise adherence will be assessed as the ability of the subjects to complete the 8-week protocol as prescribed, or whether adjustments such as reduced loads, reduced overall number of sets/reps completed, and/or days missed due to pain, soreness, or injury are needed. Finally, subjects with higher pre-participation fitness and training experience should acclimate more easily to Flywheel-only exercise (i.e., higher prescription adherence) but are predicted to also experience smaller changes in fitness than less-fit subjects. In summary, understanding the adaptive response to exercise on this unique exercise system will inform NASA Human System Risk recommendations for Lunar design reference missions (DRMs), as well as exercise capability needs for future missions and vehicles. Near-term answers on OFW efficacy and utility are critical due to the expanding reliance of exploration programs on this custom device and the approach of crewed Artemis missions.
Rationale for HRP Directed Research:	Highly constrained. Implementation of this training study will leverage planned or existing human-in-the-loop testing of the Orion flywheel device at University of Houston-Clear Lake to minimize impact on resources and budget. This study requires close coordination with internal NASA stakeholders and Flywheel project engineers. Thus, it is highly constrained and not well suited for acquisition via the normal solicitation processes.
Research Impact/Earth Benefits:	The Orion Flywheel is currently the sole exercise countermeasure system that is planned to be available on early Artemis missions. Testing the efficacy of the Orion Flywheel for protecting physical performance is necessary to ensure that the exercise device provides adequate aerobic and muscular stimuli over an extended-use timeframe. Understanding the adaptive response to exercise on this novel system will inform risk assessments for early Artemis missions, as well as exercise capability needs for future missions and vehicles. Near-term answers on Orion Flywheel efficacy will have direct mission impacts due to the expanding duration of use/reliance of exploration programs on this custom device and the approach of crewed Artemis missions. This study will also be the first to employ flywheel-type exercise as the sole modality for both aerobic and resistance exercise training 6 days per week to study the longer-term adaptive response to this exercise modality. Findings, therefore will be directly translatable to the terrestrial human health and performance field.
Task Progress:	At the beginning of 2025, the study team submitted and received approval for the Institutional Review Board (IRB) from the University of Houston-Clear Lake (UHCL), the institute of study execution. Associated test plans and study protocols have also been developed. However, over the previous reporting period, there has been a large rescoping of the study design due to a reduction of the study's budget. Specifically, the study arms were reduced from 3 exercising cohorts and 1 control cohort to a single Orion Flywheel exercise group. The majority of the originally proposed fitness and task performance measures were preserved at pre-, mid-, and post-participation timepoints, although blood biomarkers were largely reduced to those focused on inflammation and muscle damage. Additionally, due to the dynamic nature of flight hardware development schedules, there have been further delays to the delivery of the Flywheel exercise device to the study site. Current expectations are to initiate the exercise intervention in Spring 2026. Finalization of study documentation and IRB updates are currently underway.

Bibliography Type:	Description: (Last Updated:)